

An External Electret Microphone for AudioMoth

theteam@openacousticdevices.info

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AudioMoth recordings typically use the built-in sensitive MEMS microphone. However, external electret microphones are also supported through the 3.5 mm audio socket that is already fitted to the AudioMoth Dev and AudioMoth USB Microphone boards, and that can be added to the standard AudioMoth device. This application note describes an external electret microphone designed specifically for AudioMoth.

1 External Microphone

The AudioMoth external microphone uses a sensitive Primo EM419N electret capsule¹. This capsule is a new replacement for the older, popular, and now discontinued, Primo EM258 capsule. The Primo EM419N capsule is encapsulated in a small plastic housing, custom designed by Primo for Open Acoustic Devices, and is fitted with a short 15 cm cable terminating in a 3.5 mm mono jack plug (see Figure 1).

The external microphone fits within a custom injection-moulded microphone holder with an M16 thread, nut and washer. It is sealed with a replaceable Gore GAW334 acoustic vent to provide full immersion waterproofness². The microphone is supplied with two spare acoustic vents and a windshield³, and fits directly into the existing hole in the base of the AudioMoth Dev case⁴ (see Figure 4).

2 Frequency Response

Electret microphones typically exhibit low self-noise and excellent sensitivity to low-frequency sounds. Figure 5 shows the datasheet of the Primo EM419N capsule, including its manufacturer-tested frequency re-



Figure 1: AudioMoth external microphone showing the microphone cable with 3.5 mm mono plug, waterproof mount, and windshield.

sponse curve. For comparison, Figure 6 shows the relevant datasheet pages of the Knowles SPU0410LR5H-QB MEMS microphone that AudioMoth uses⁵.

Direct comparison of the datasheet sensitivity and frequency response is complicated by the performance of the internal AudioMoth circuitry (shown in Figure 3). This is particularly so at low frequencies where the frequency response of the bias and amplifier feedback circuits becomes more significant.

Thus, Figure 2 shows a direct measured comparison of the sensitivity of the external electret microphone, relative to the internal MEMS microphone, from 20 Hz

¹<https://primomic.com/pdf/EM419N.pdf>

²<https://www.gore.com/sites/default/files/resources/pdf/2025-09/gore-acoustic-vents-portfolio-datasheet-en.pdf>

³<https://www.openacousticdevices.info/product-page/audiomoth-external-microphone>

⁴<https://www.openacousticdevices.info/product-page/audiomoth-dev-case>

⁵https://media.digikey.com/pdf/Data%20Sheets/Knowles%20Acoustics%20PDFs/SPU0410LR5H-QB_RevH_3-27-13.pdf

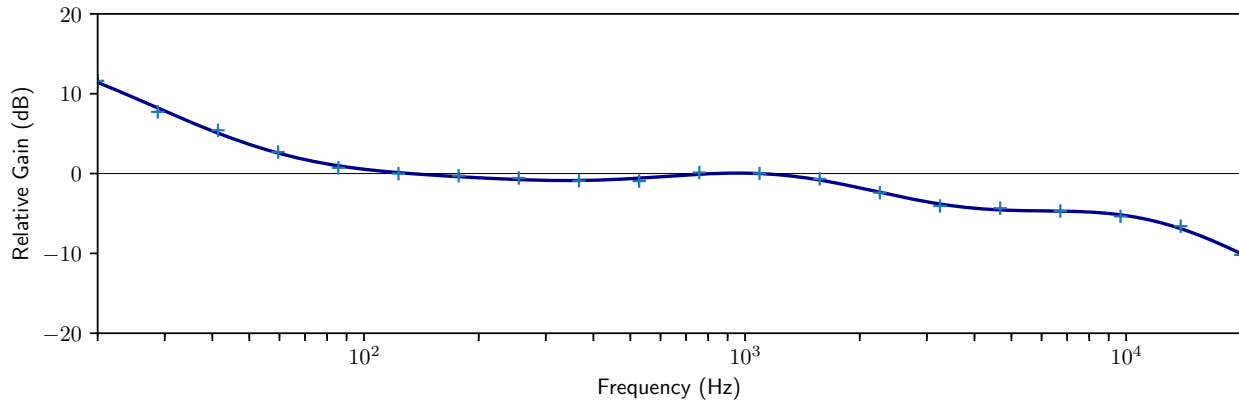


Figure 2: Comparison of measured Primo EM491N electret capsule frequency response, relative to the Knowles SPU0410LR5H-QB MEMS microphone, from 20 Hz to 20 kHz.

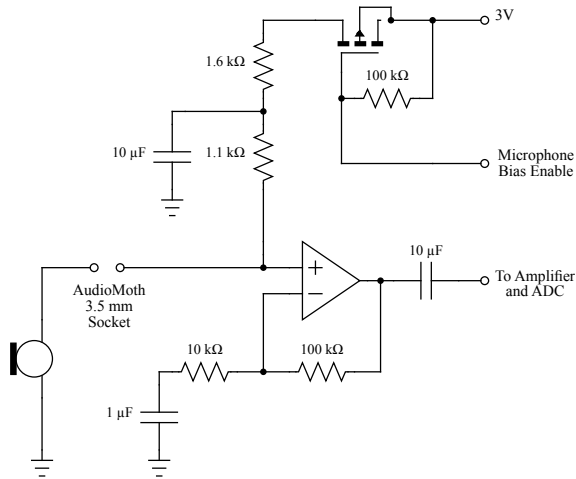


Figure 3: AudioMoth external microphone circuit, incorporating a Maxim MAX4466 microphone preamplifier integrated circuit and software-controlled microphone bias circuit, connected to the external microphone.

to 20 kHz. This measurement was performed by recording broad spectrum white noise, played from a speaker at 70 dB SPL at 1 m distance, to two AudioMoth, one using the internal MEMS microphone, and one using the external electret microphone described here.

3 Discussion

The external Primo EM419N microphone exhibits better low frequency sensitivity, approximately the same sensitivity around 1 kHz, and poorer high frequency sensitivity compared to the internal MEMS microphone. The internal MEMS microphone is preferred for recording bats due to its excellent high-frequency sensitivity which is measured by the manufacturers to 80 kHz.



Figure 4: AudioMoth external microphone, and dead cat wind shield, fitted to the existing hole in the base of the AudioMoth Dev case.

However, the external electret capsule excels for recording low-frequency sounds. To date, it has demonstrated excellent performance in our own deployments recording the low-frequency rumbles of forest elephants and the low-frequency booms of the Bittern (*Botaurus stellaris*), alongside making high-quality recordings of bird song at higher frequencies. Furthermore, its robust design and ease of mounting makes it suitable for deployments of AudioMoth boards within custom housings.

エレクトレットコンデンサマイクロホン
Electret Condenser Microphone
EM419N

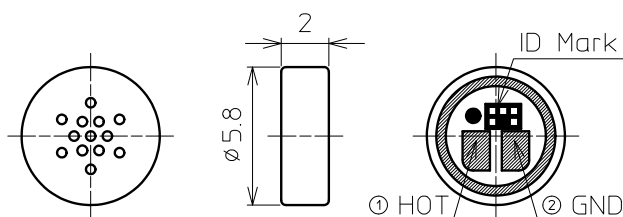
RoHS2 **1. 特長 Features**

超高感度、高 S/N 比、小型薄型、軽量.

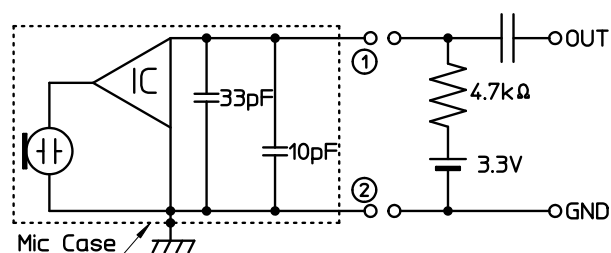
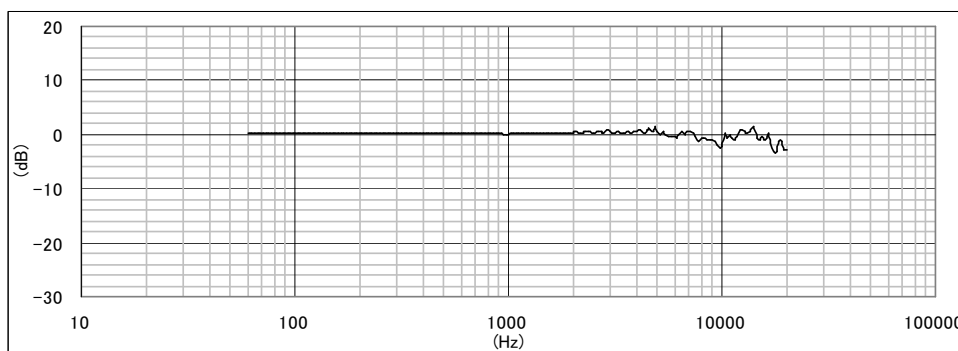
Ultra High sensitivity, High S/N Ratio, Small size, Thin & light weight.

2. 仕様 Specifications

2-1) 指向性	Directional Characteristic	: 無指向性 Omni-directional
2-2) 感度	Sensitivity	: -29 dB $\pm$ 3dB at 1 kHz (0 dB = 1 V/Pa) VCC = 3.3 V, RL = 4.7 k Ω
2-3) インピーダンス	Impedance	: 2.3 k Ω $\pm$ 30 % at 1 kHz
2-4) 位相	Phase	: 5 deg $\pm$ 10 deg at 30 Hz
2-5) S/N 比	S/N Ratio	: 75 dB at 1 kHz (1 Pa, A weighted network)
2-6) 動作電圧	Operating Voltage	: 3.3 V
2-7) 消費電流	Current Consumption	: 400 μ A Max
2-8) 最大入力音圧	Max Input Sound Pressure Level	: 120 dB S.P.L.
2-9) 重量	Weight	: 0.13 g

3. 外観図 Appearance Drawing [mm]

※仕様によって識別マークは異なります。
ID Mark depends on specification.

4. 回路図 Circuit Diagram**5. 周波数特性 Frequency Response**株式会社 **PRIMO**

〒 190-1232 東京都西多摩郡瑞穂町長岡 2-3-5

PRIMO CO., LTD.

2-3-5 Nagaoka, Mizuho-Machi, Nishitama-Gun, Tokyo 190-1232, Japan

Tel. 042-556-8711 (営業課) +81-42-556-8711 (Sales dept.) Fax. 042-556-8721

e-mail : eigyo@primocorp.co.jp

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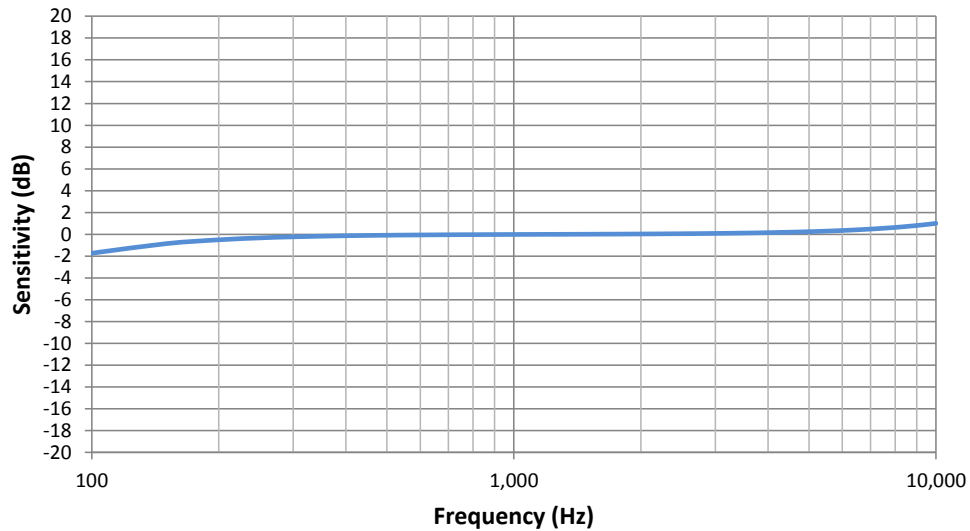
Figure 5: Primo EM419N capsule datasheet.



SPU0410LR5H-QB

3. FREQUENCY RESPONSE CURVE

Typical Free Field Response
Normalized to 1kHz



Preliminary Ultrasonic Free Field
Response Normalized to 1kHz

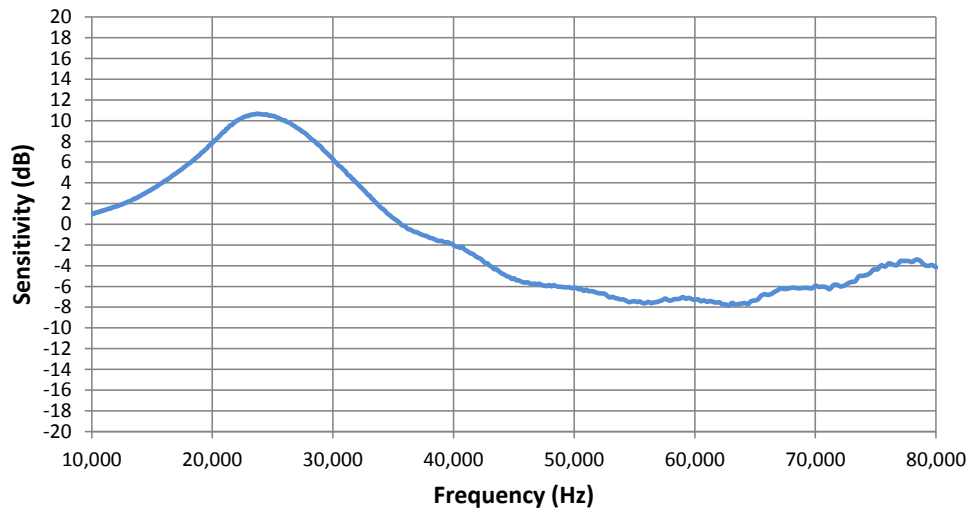


Figure 6: Knowles SPU0410LR5H-QB frequency response plot from Knowles datasheet.